

INGENIOUS SIGNAL GENERATION Driving Industry Progress

As telecommunications, defence, and automotive industries increasingly adopt advanced electronic testing, signal generators, particularly super high frequency and vector models, are leading the technological shift.

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Signal generators are key for testing and maintaining electronic devices in the telecommunications, defence, and automotive industries. The market, worth \$1.39 billion in 2023, is projected to reach \$1.98 billion by 2028. This growth is fuelled by advances in telecommunications, aerospace, defence, and the integration of IoT and 5G technologies. Innovations like the super high frequency signal generator (SHFSG) are pushing the market towards specialised applications. Agilent Technologies and Ametek Inc. are leading this trend, especially in Asia-Pacific.

The significant advancements in the electronic testing and signal generation field were driven by the introduction of vector signal generators and other sophisticated tools, enhancing efficiency, accuracy, and usability. The N5186A MXG, part of Keysight's X-Series signal generator lineup, represents the next generation of high-performance vector signal generators (VSGs). It delivers multiple, individually complex signals essential for dense wideband multi-channel applications. These



Anritsu MG36271A signal generator
(Credit: Anritsu)



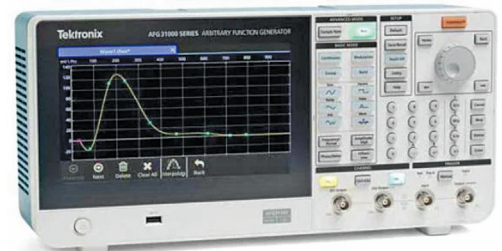
Keysight N5186A MXG signal generator
(Credit: Keysight)



Rohde & Schwarz SMB100B microwave signal generator (Credit: R&S)



Rigol DG1022Z signal generator (Credit: Rigol)



Tektronix AFG31000 series arbitrary function generator (Credit: Tektronix)

technologies, boosted by better digital-to-analogue conversion, increased processing power, and advanced software algorithms, have made signal generators more integrated, user-friendly, and versatile. Incorporating AI and IoT has further transformed their functionality, enabling automated, interconnected testing environments crucial for supporting the rapid development cycles in modern electronic industries.

Signal generation software and enhanced programmability

Signal generation software offers comprehensive tools for creating a wide range of signals, including 5G NR, WLAN, and custom modulation, ensuring compliance with the latest industry standards. This software accelerates signal creation and reduces simulation time with interactive controls and graphical navigation, making it highly useful across various industries such as cellular, aerospace, and broadcasting.

New signal generator models feature high dynamic ranges, achieving up to an 80dB programmable range. They offer low-power options suitable for diverse applications and enhanced programmability for various testing scenarios. These